## Questions

**Q1.**

Figure 5 shows a PET scanner used to detect cancerous tumours.

A radioactive isotope is injected into a patient.

The isotope is absorbed by the tumour.

The isotope emits positrons from the location of the tumour.

The ring of radiation detectors rotates around the person.

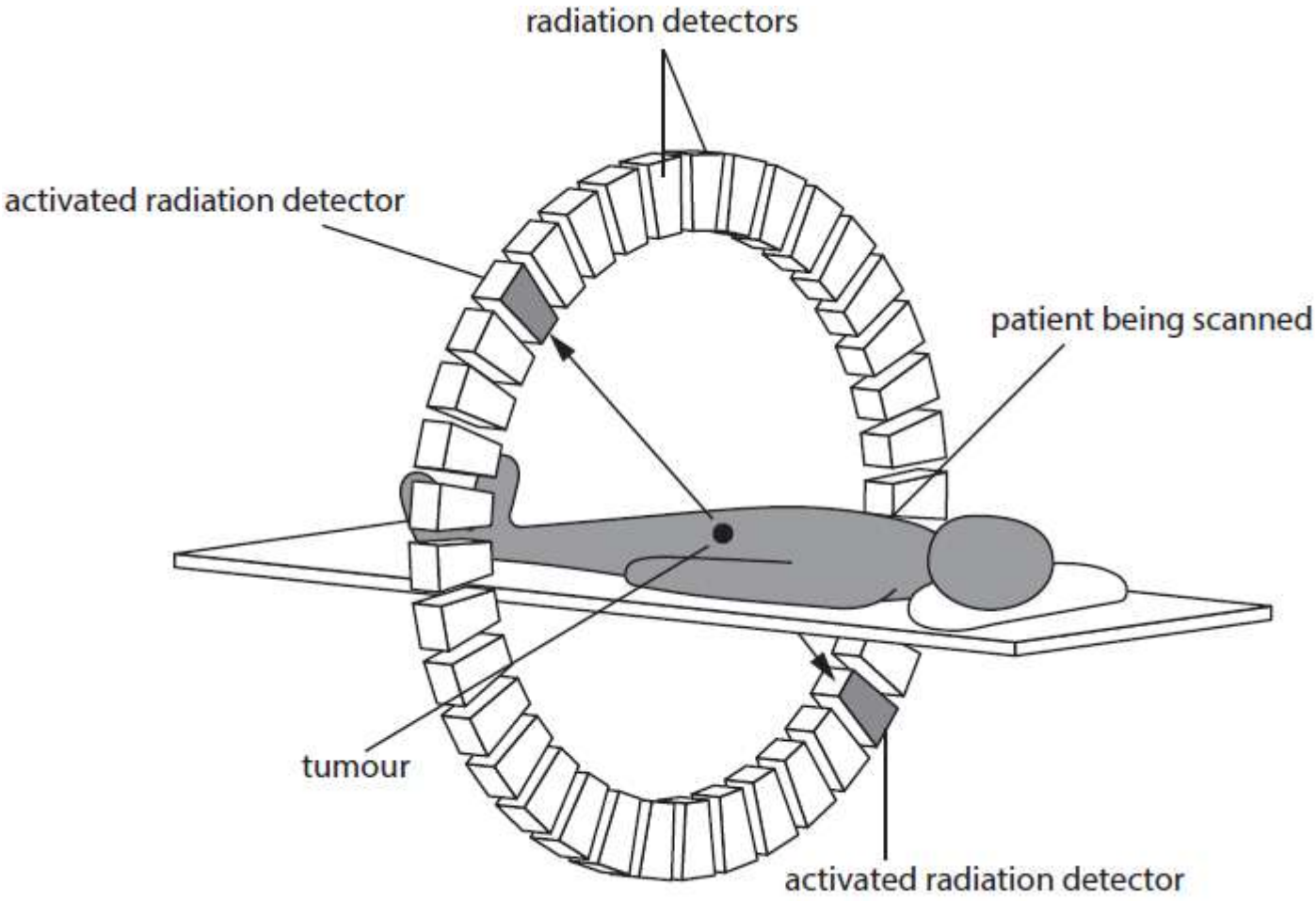


**Figure 5**

(i) Explain how the scan can give the location of the tumour.

**(3)**

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(ii) Explain why the radioactive isotope injected into the patient must be produced near to the place where it is to be used.

**(2)**

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**(Total for question = 5 marks)**

**Q2.**

A teacher determines the background radiation count rate in a laboratory.

Explain how to determine a value for the background radiation count rate.

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**(Total for question = 3 marks)**

**Q3.**

Early in the twentieth century, scientists fired a beam of alpha particles at thin gold foil.

Figure 2 shows the main parts of their experiment with some results.

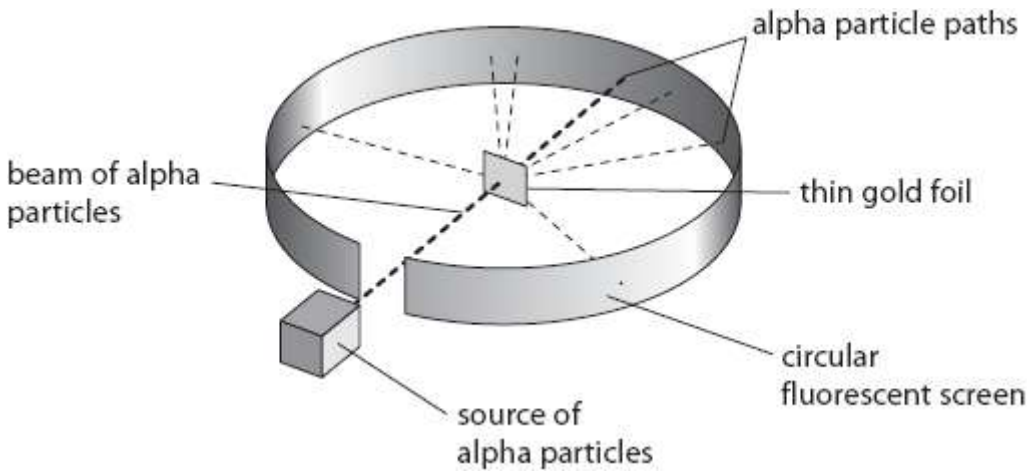


**Figure 2**

Explain how the results of the experiment shown in Figure 2 support the nuclear model of an atom.

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**(Total for question = 3 marks)**

**Q4.**

Explain how neutrons enable a nuclear chain reaction to take place.

**(2)**

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**(Total for question = 2 marks)**

**Q5.**

Some isotopes are unstable.
They emit $\beta^-$ particles when they decay.

Explain how a nucleus changes when a $\beta^-$ particle is emitted.

**(2)**

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**(Total for question = 2 marks)**

**Q6.**

The apparatus that was used in the experiment is shown in Figure 10.

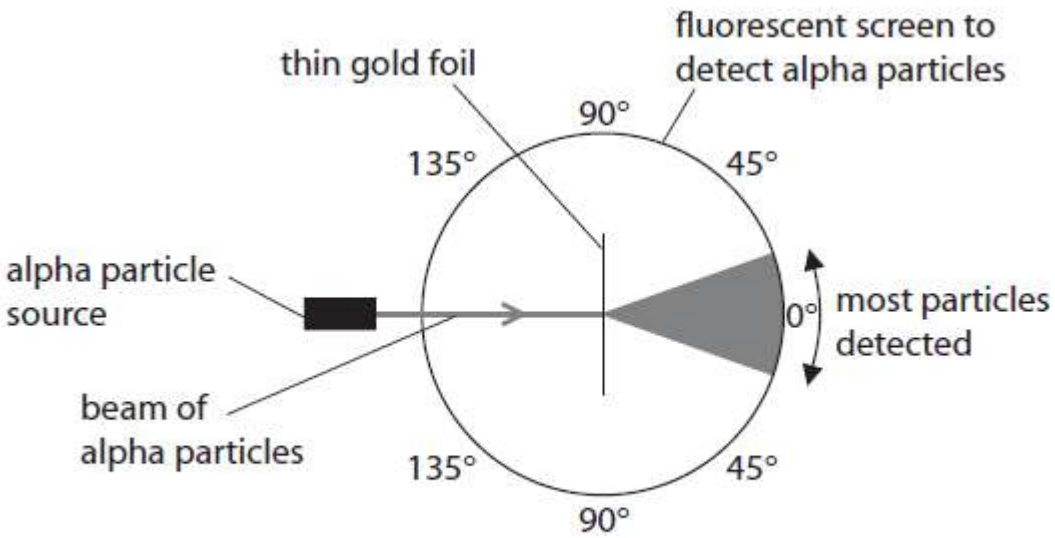


**Figure 10**

(i) The number of particles detected at each angle in a given time is shown on the graph in Figure 11.

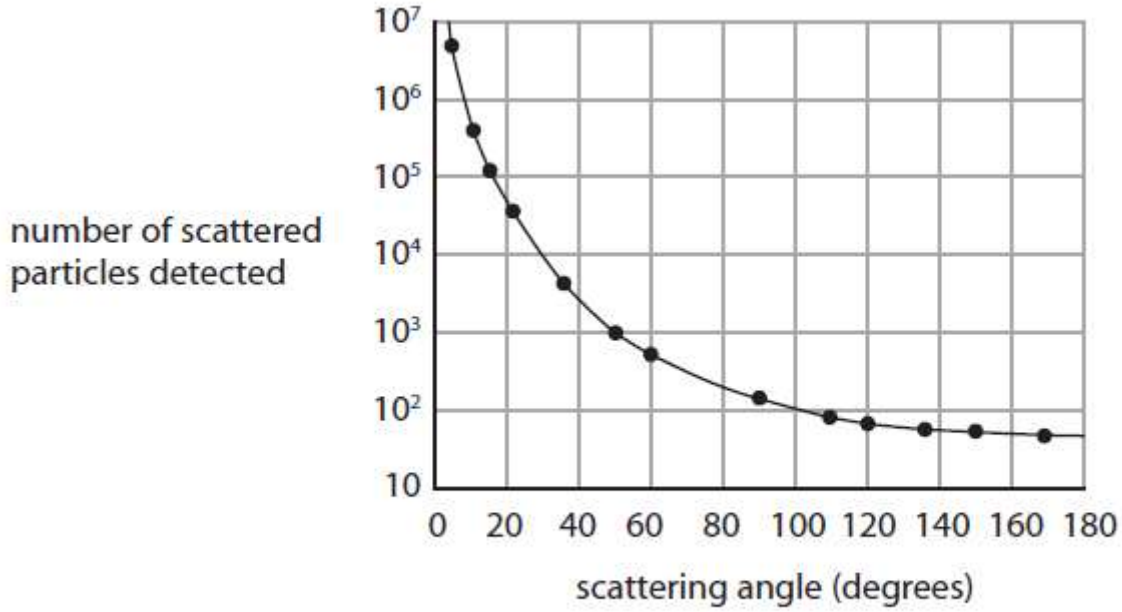


**Figure 11**

Use information from the graph.
Estimate the ratio of the number of particles scattered through 5° to the number of particles scattered through 100°.

**(2)**

ratio = .....

(ii) Explain how the difference in the number of particles scattered at different angles gives evidence for the current model of the structure of the atom.

**(4)**

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**(Total for question = 6 marks)**

**Q7.**

Figure 6 is a diagram of a nuclear reactor.

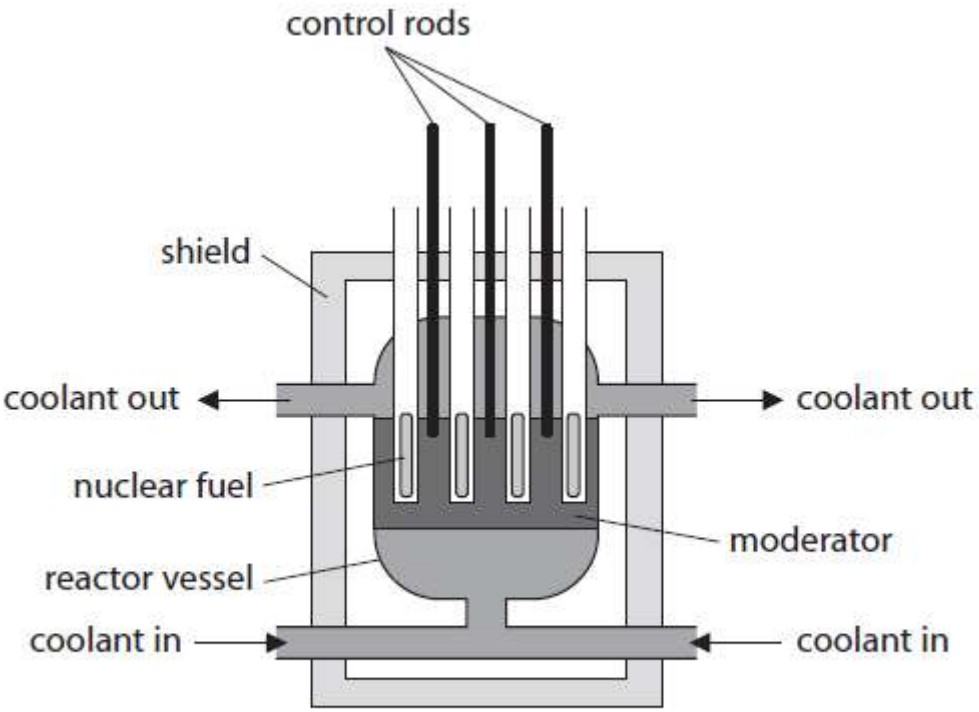


**Figure 6**

(i) Explain how pushing the control rods further into the reactor slows down the nuclear chain reaction.

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(ii) The moderator in a nuclear reactor slows down the neutrons so that the neutrons are more likely to start other fission reactions.

In a nuclear reactor,

- the average speed of the fast neutrons is 3.0 × 107 m/s
- the average speed of the slow neutrons is 4.0 × 103 m/sCalculate the average speed of the slow neutrons as a percentage of the average speed of the fast neutrons.

**(2)**

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(iii) The nuclear reaction is the first stage in the process of generating electricity.

Describe how energy is transferred from the nuclear reaction to the next stage in the process.

**(2)**

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**(Total for question = 6 marks)**

**Q8.**

In a nuclear reactor, a chain reaction is produced and controlled.

Explain what happen inside a nuclear reactor if neutron speeds are not controlled.

**(3)**

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**(Total for question = 3 marks)**

**Q9.**

Explain the use of a moderator in a nuclear reactor.

**(3)**

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**(Total for question = 3 marks)**

**Q10.**

Carbon–14 decays into nitrogen–14.

The symbol for nitrogen–14 is $^{14}_{7}N$

Explain what happens in a carbon-14 nucleus when it decays to a nitrogen-14 nucleus.

**(2)**

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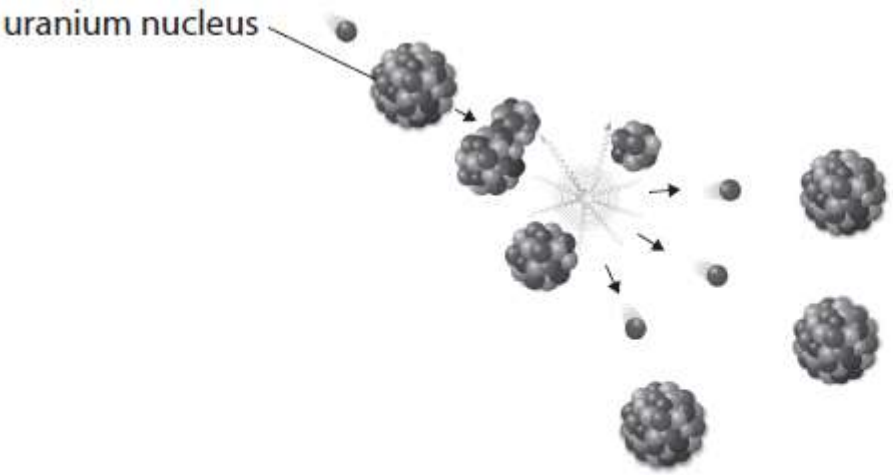
**(Total for question = 2 marks)**

**Q11.**

Figure 8 shows what happens during a fission chain reaction.




(Source: © Designua/Shutterstock)

**Figure 8**

Explain what happens when **one** neutron interacts with **one** uranium nucleus.

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**(Total for question = 3 marks)**

## Mark Scheme

Q1.

| Question | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | explanation linking any 3 from:-<br><br>positrons and electrons annihilate (1)<br><br>(two) gamma (rays) produced/emitted (1)<br><br>in opposite directions /at 180° (1)<br><br>detected by radiation detector/ gamma cameras/scintillation counters (1)<br><br>at (almost) the same time (1)<br><br>time difference gives distance difference (1) | ignore positrons for this marking point<br><br>allow triangulation | 3<br>AO2.2 |

| Question | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | explanation linking any two from<br><br>must be used a short time after production (1)<br><br>half-life is short (1) | must be used while activity is high | 2<br>AO1.2 |
| | activity decreases rapidly/decays rapidly (1) | accept decays before use / does not last long | |

**Q2.**

| Question | Answer | Additional guidance | Mark |
|---|---|---|---|
| | an explanation linking three from<br><br>use of G-M tube (with counter) (1)<br><br>no (obvious) radioactive sources present (1)<br><br>measure (number of) counts in a given time (1)<br><br>divide number of counts by time (1)<br><br>repeat readings (1)<br><br>calculate the average value (1) | allow Geiger counter / rate meter<br><br>allow measure count rate / activity if rate meter used<br><br>take readings in different positions in laboratory | 3<br>AO3.3 |

**Q3.**

| Question number | Indicative content | Mark |
|---|---|---|
| | An explanation that combines identification via a judgment (2 marks) to reach a conclusion via justification/reasoning (1 mark):<br><br>• some alpha particles go straight through (1)<br>• some alpha particles scattered (1)<br><br>• idea of all mass / (positive) charge concentrated in centre /nucleus (1)<br>• mainly empty space (in rest of atom) (1) | (3) |

**Q4.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation linking two of<br><br>a neutron collides with/fired at a (uranium) nucleus (1)<br><br>releasing (2/3) extra neutron(s) (1)<br><br>which go on to collide with more nuclei (1) | <br><br>neutron absorbed by nucleus.<br>accept atom for nucleus<br><br><br>accept atom for nucleus | (2) |

**Q5.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation that combines understanding (1 mark) and reasoning (1 mark) linking:<br><br>• number of neutrons decreases by one (1)<br>• number of protons increases by one.(1) | <br><br>a neutron becomes a proton plus an electron for (2) marks | (2) |

**Q6.**

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | reading from graph (1)<br><br>(at 5 degrees:) number between $10^6$ and $10^7$<br>**AND**<br>(at 100 degrees:) $10^2$<br><br>evaluation (1)<br><br>$\frac{\text{number between } 10^6 \text{ and } 10^7}{10^2}$<br><br>**OR**<br><br>between $10^4$:1 and $10^5$:1<br>between 10 000:1 and 100 000:1<br><br>**OR**<br><br>between $10^4$ and $10^5$<br>between 10 000 and 100 000 | <br><br>(e.g. $10^{6.5}$)<br><br><br><br><br><br><br><br><br><br>(e.g. $10^{4.5}$:1 or $10^7$:$10^2$)<br>allow any correct ratio not in its simplest form<br><br><br>(e.g. $10^{4.5}$)<br><br>award full marks for correct answer without working<br><br>inverted ratio scores 1 mark maximum | (2)<br>AO2 |

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | an explanation including any **four** from:<br><br>Observations<br><br>most (alpha particles) pass (straight) through the foil (with little deflection) (1)<br><br>some (alpha particles) are {scattered / deflected} through {small angles / less than 90 degrees} (1)<br><br>(very) few (alpha particles) are {scattered / deflected} through {large angles / greater than 90 degrees} (1)<br><br>Conclusions<br><br>atoms are mainly empty space (1)<br><br>there must be a nucleus / something inside the atom (1)<br><br>(nucleus / something) must be {small / heavy / dense / concentrated / charged / positive} (1) | ignore electrons<br><br>ignore refracted<br>allow repelled<br><br>allow rebound / reflect / back scattering / bounce back<br><br>ignore electrons | **(4)**<br>**A01,AO3** |

**Q7.**

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | an explanation linking:<br><br>(control rods) absorb neutrons (1)<br><br>(so) fewer (neutrons) available for chain reaction (1) | ignore slow down the nuclear chain reaction<br><br>accept (control rods) block neutrons<br>accept neutrons can't pass through (control rods)<br><br>fewer fission(s) (reactions) | (2)<br>AO1 |

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | $\frac{4(.0 \times 10^3)(\times 100)}{3(.0 \times 10^7)}$ (1)<br><br>$1.3 \times 10^{-2}$ (%) (1) | <br><br>0. 013 (%)<br><br>allow 0.01 (%)<br><br>power of ten error scores 1 mark maximum<br><br>award full marks for the correct answer without working | (2)<br>AO2 |

| | Answer | Additional guidance | Mark |
|---|---|---|---|
| (iii) | A description to include:<br><br>(from) kinetic energy (of fission fragments) (1)<br><br>(transferred to) thermal energy (of coolant) (1) | accept references to energy stores<br><br>accept energy in nuclear store<br>accept nuclear energy / gamma radiation energy / binding energy / mass<br><br>(to) thermal store (in coolant)<br><br>accept heat for thermal<br><br>if no other marks awarded allow steam transfers thermal energy/heat from reactor to kinetic energy of turbine for 2 marks | (2)<br>AO1 |

**Q8.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation that combines identification – knowledge (1 mark) and reasoning/justification – knowledge (2 marks):<br><br>• reaction will slow down (1)<br>• because there are fewer fissions (1)<br>• because fission more likely with slow neutrons (1) | allow<br><br>reactor shuts down/eq<br><br>fission requires slow neutrons<br>thermal neutrons for slow neutrons | (3) |

**Q9.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation that combines understanding (1 mark) and reasoning/justification - understanding (2 marks):<br><br>• slows down neutrons (1)<br><br>• increase chance of collision / fission (1)<br>• maintain the reaction rate (1) | <br><br><br><br>to allow capture by nucleus<br><br>increase the reaction rate | **(3)** |

**Q10.**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | An explanation linking: | | **(2)** |
| | neutron (decays) to proton (1) | mass number stays the same but atomic number increases by 1<br><br>accept answers in terms of quarks<br><br>(dud becomes uud) | |
| | beta emitted (1) | beta decay / β seen<br><br>NOT $\beta^+$/beta plus<br><br>allow (fast) electron emitted<br><br>allow for 2 marks:<br><br>n □ p + e<br><br>OR<br><br>$^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}\beta^{(-)}$ | |

**Q11.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| | an explanation linking any **three** from | | **(3)**<br>**AO1.1** |
| | neutron absorbed (into uranium) (1) | allow collides/hits/joins with | |
| | (causing) nucleus becomes unstable (1) | do not allow atom for nucleus | |
| | (so nucleus) breaks up/splits (into two other nuclei) (1) | | |
| | (then) releasing (more) neutron(s) (1) | | |
| | releases gamma rays / energy (1) | | |
| | | if no other marks scored award 1 mark for neutrons go on to hit other nuclei | |